

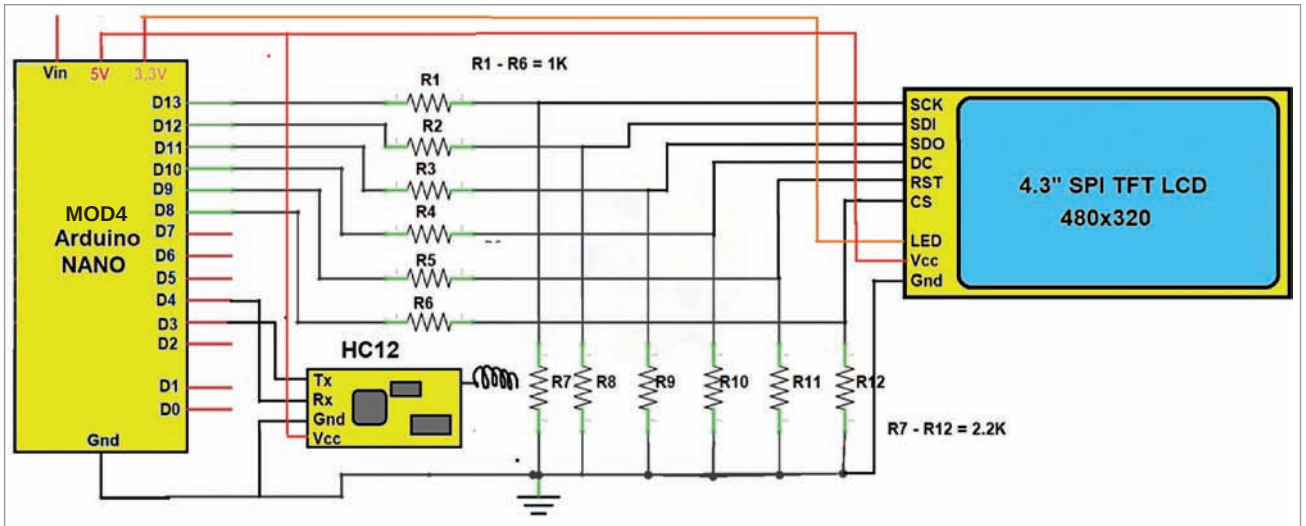


Fig. 6: Circuit diagram of the RDMS receiver

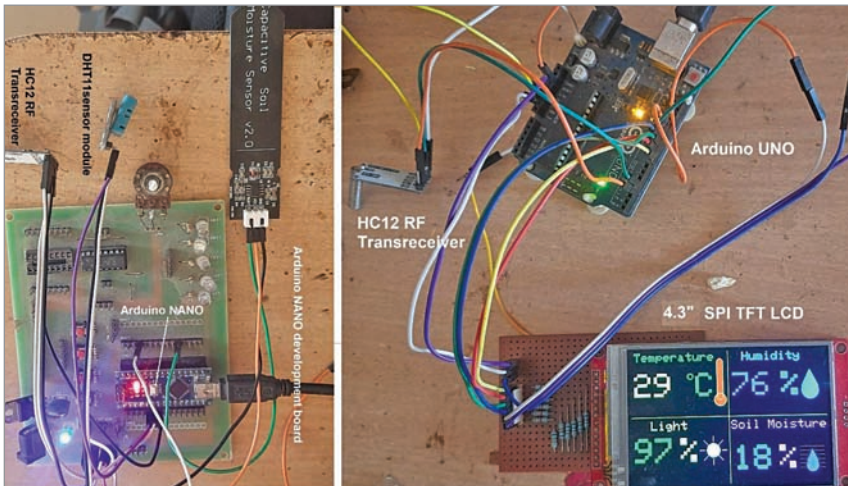


Fig. 7: Author's prototype

MOD4) as the MCU and 434MHz RF transceiver modules (HC12). The block diagram of the receiver and TFT LCD monitor is shown in Fig. 5.

Circuit diagram of the RDMS receiver. The circuit for the RDMS data receiver (Fig. 6) comprises an RF transceiver module, an Arduino Nano board (MOD4), and a TFT LCD monitor.

First, connect a 5V power supply to MOD4 after uploading the source code. Then, connect all components according to Fig. 6. The HC12 module has four pins, and its connections to MOD4 mirror those in the

transmitter circuit. The TFT LCD monitor includes nine interfacing pins, which are connected to MOD4 as detailed in Table 2.

A voltage divider network, built using 1-kilo-ohm and 2.2-kilo-ohm resistors, is connected between MOD4 and the TFT LCD pins (Fig. 6). As the TFT LCD operates at 3.3V and MOD4 pins output 5V, this network converts the 5V output to 3.3V, ensuring proper operation.

Circuit operation

Initially, both the RDMS transmitter and receiver are off. When the

TABLE 2
PIN CONNECTIONS
BETWEEN THE TFT LCD
MONITOR AND MOD4

TFT LCD Pin	Arduino Nano Pin
VCC	5V
GND	GND
LED	3.3V
CS	D8
RST	D9
DS	D10
SDO	D11
SDI	D12
SCK	D13

receiver is powered on, the TFT LCD monitor activates. Once the transmitter is switched on, MOD1 starts transmitting data.

In the transmitter, MOD1 reads the digital output of the DHT11 sensor via the D2 pin, which provides temperature and humidity values. The soil moisture sensor generates an analogue voltage output between 0 and 5V, corresponding to soil moisture content. MOD1 uses its internal ADC to convert this analogue signal into a digital value between 0 and 1023, which is then mapped to a 0-100% scale for soil moisture measurement. Similarly, as the light